Xiaofeng Yang

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Education

Clemson University — Clemson, SC
Ph.D. in Biomedical Data Science and Informatics GPA: 3.9 / 4.0 — *Aug 2024 – Present*

New York University — New York, NY
M.S. Biostatistics GPA: 3.82 / 4.0 — *Sep 2021 – May 2023*

University of Iowa — Iowa City, IA
B.S. Mathematics & Statistics (Dual Degree) GPA: 3.7 / 4.0 — *Aug 2017 – May 2021*

Research & Projects

Deep Learning Benchmarking: Multiple Instance Learning at Scale — Aug 2024 – Jul 2025
Clemson University | Research Assistant — *Clemson, SC*
- Designed and implemented a benchmarking framework evaluating **10 MIL algorithms** across 6 training configurations on 6 datasets from 4 imaging modalities (CT, X-ray, MRI, Ultrasound)
- Evaluated models on **AUROC**, calibration (**Brier score**), and sample efficiency via k-fold cross-validation; dot-product attention was the most consistent top performer across all datasets

End-to-End Multi-Stage Computer Vision Inference Pipeline — Aug 2024 – Jan 2026
Clemson University | Research Assistant — *Clemson, SC*
- Architected a **three-stage sequential inference pipeline** using ResNet-50 with attention-based MIL aggregation: modality classification, relevance filtering, and binary classification from video clips
- Achieved **AUROC 0.89** on internal test set; generalized to external dataset from a different institution and hardware vendor with AUROC 0.89; error analysis via **UMAP** and confusion matrices

Multi-Task Learning for Multi-Label X-ray Classification — Aug 2024 – Jan 2026
Clemson University | Research Assistant — *Clemson, SC*
- Built end-to-end data pipeline for 838 patient studies (3,727 images); designed **Single-Task** and **Multi-Task MIL** architectures using pretrained **ResNet-50d** with task-specific attention pooling heads
- Multi-Task model outperformed Single-Task on 8 of 9 targets; **ROC-AUC 0.80–0.94**; led all evaluation and visualization including ROC curves and cross-task performance plots

Self-Supervised Learning with Equivariance Regularization — Jan 2025 – Present
Clemson University | Research Assistant — *Clemson, SC*
- Extended **VICReg** with a novel **equivariance loss** enforcing geometric consistency between augmented view pairs, adding to existing variance-invariance-covariance objectives
- Implemented full training framework with **multi-GPU distributed training** (PyTorch DDP) supporting ResNet-50/18 encoders for scalable pretraining on large unlabeled image datasets

EEG Signal Processing & Statistical Analysis — Oct 2022 – May 2023
NYU Langone Health | Research Assistant — *New York, NY*
- Processed large-scale **EEG time-series data** using MNE-Python; applied **ICA** for signal decomposition and artifact removal; conducted multivariate statistical analysis for a prospective cohort study

Publications

P1. Formby, B.*, **Yang, X.***, Wang, K.C., Tsai, H.L., & Smith, D.H. "Bridging Clinical Diagnosis and Deep Learning Workflows: An Analysis of Multiple Instance Learning for Exam-Level Classification." *IEEE Journal of Biomedical and Health Informatics*, under review. *(* co-first authors)*

P2. Gearhart, A.*, **Yang, X.***, Lai, W., Mayourian, J., Powell, A.J., Smith, D.H., & Baker, G.H. "Towards a Fully Automated Deep Learning Echocardiography Pipeline for Patent Ductus Arteriosus Detection in Preterm Infants." *Journal of Ultrasound in Medicine*, submitted. *(* co-first authors)*

P3. Floyd, S.B., Cordero Romero, M., **Yang, X.**, Smith, D.H., Pill, S., Sridhar, M., & Jeray, K. "Evaluation of a Computer Vision Model to Identify Proximal Humerus Fracture Features." In preparation.

P4. Gyasi, F.*, Scherbakov, D.*, Sun, Y., **Yang, X.**, Heider, P., Fronk, G., Ridings, L., Alekseyenko, A.V., & Lenert, L.A. "Using Real-World Data on Depression from EHR-based Research Networks: A Scoping Review." *Journal of Clinical and Translational Science*, under review.

Skills

Languages: Python, R, SQL, MATLAB

ML/DL Frameworks: PyTorch, torchvision, scikit-learn, HuggingFace

Computer Vision: OpenCV, ResNet, Vision Transformer (ViT), Attention Mechanisms